Rare-miss growth under sliding-window constraints

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Abstract

We consider products of two fixed strictly positive matrices under the rule that every window of k factors contains at most m occurrences of a designated miss mode. With dimension and m fixed, we identify the first-order correction to the logarithmic worst-case growth rate as k tends to infinity. The coefficient is the logarithm of the largest Perron-weighted gain of a cluster containing at most m misses. A uniform cluster-packing argument gives the upper bound, and periodic products give the matching lower bound. We also give a finite-search enclosure for the coefficient and examples showing that optimal pair gaps cannot be restricted to a dimension-independent finite list. A rank-one reset example illustrates why period-k witnesses need not be optimal at finite k.

Draft status. Priority assessment is incomplete. The underlying AI-assisted arguments have received internal automated mathematical review. The main cluster law now has a kernel-checked Lean proof in the accompanying repository. The additional propositions have not all been formalized. External specialist review and author metadata remain pending.

1 Introduction and scope

Missed updates in a discrete-time system can be represented by a restricted switching sequence. A sliding-window rule limits the number of misses in every prescribed number of slots. It is stronger than a restriction on asymptotic average frequency, and it still permits short bursts. Constrained switching has exact spectral and multinorm descriptions [4, 7]; weakly-hard control also admits useful scalar convergence-rate abstractions [2].

Here the window size varies while the matrices and miss budget remain fixed. We determine the coefficient of the resulting k^{-1} correction for strictly positive modes. The main issue is an upper bound uniform over *all* admissible words. Evaluating a collection of periodic words provides lower bounds but does not settle this uniform comparison.

Rank-one product identities and maximum-cycle methods are established [1, 5]. Perron pairings also occur in first-order slow-switching expansions for periodic and random environments [6]. The latter hold in a regime that stretches retained environmental intervals with fixed occupation proportions. Our miss clusters can have fixed duration, and the optimization ranges over a hard sliding-window language. Hybrid singular-perturbation theory provides further related reduced-system limits [3], but its timing hypotheses do not directly supply the packing comparison used here.

The result below concerns fixed strictly positive discrete matrices. It is not a theorem for arbitrary signed modes, growing miss budgets, continuous intersample safety, or nonlinear plants. Nor does it assert that every matrix pair is realizable by a particular feedback/dropout architecture. The exact reset calculation in Section 6 is presented as an application of established rank-one

machinery. No first-in-literature claim is made in this draft; the remaining review obligations are recorded in Section 7.

2 The cluster law

Fix integers $d, m \geq 1$ and entrywise strictly positive matrices $C, D \in \mathbb{R}^{d\times d}$. For integers $k > m$, let

$$\Sigma_{m,k} = \left\{ \sigma \in \{0,1\}^{\mathbb{N}_0} : \sum_{i=t}^{t+k-1} (1-\sigma_i) \leq m \text{ for every } t \in \mathbb{N}_0 \right\}.$$

The symbol 1 denotes success and 0 denotes a miss. Write $A_1 = C$, $A_0 = D$, and define the constrained growth rate

$$\rho_{m,k} = \limsup_{N\to\infty} \sup_{\sigma\in\Sigma_{m,k}} \left\| A_{\sigma_{N-1}} \cdots A_{\sigma_0} \right\|^{1/N}. \tag{1}$$

Any fixed matrix norm gives the same rate. Put

$$\lambda = \operatorname{spr}(C), \qquad B = C/\lambda, \qquad E = D/\lambda.$$

Perron–Frobenius theory gives positive vectors r, ℓ such that

$$Br = r, \qquad \ell^{\mathsf{T}} B = \ell^{\mathsf{T}}, \qquad \ell^{\mathsf{T}} r = 1, \qquad B^n \longrightarrow P := r\ell^{\mathsf{T}}. \tag{2}$$

For $j \geq 2$ and gaps $\boldsymbol{g} \in \mathbb{N}_0^{j-1}$ define

$$K_j(\boldsymbol{g}) = EB^{g_{j-1}} E \cdots B^{g_1} E, \qquad K_1 = E.$$

Thus g_1 is the number of successes after the first miss in chronological order; matrices multiply in the opposite order. Define

$$S_j = \sup_{\boldsymbol{g}\in\mathbb{N}_0^{j-1}} \ell^{\mathsf{T}} K_j(\boldsymbol{g}) r, \qquad M_m = \max\{1, S_1, \ldots, S_m\}. \tag{3}$$

For $j = 1$ the supremum has no gap variables. Each S_j is positive and finite because the powers of B are bounded and j is fixed. The supremum need not be attained at finite gaps.

Theorem 1 (Fixed-budget rare-miss law)**.** *With C, D, d, m fixed as above,*

$$\lim_{k\to\infty} k \log \frac{\rho_{m,k}}{\lambda} = \log M_m. \tag{4}$$

Equivalently, $\log \rho_{m,k} = \log\lambda + k^{-1} \log M_m + o(k^{-1})$.

The all-success sequence explains the maximum with 1 in (3). In particular, the correction is nonnegative. The theorem does not identify an exactly maximizing word at each finite window size.

3 Proof of Theorem 1

3.1 Perron separation and joining

Use the vector norm $\|x\|_r = \max_i |x_i|/r_i$ and its induced matrix norm. For every nonnegative matrix W,

$$\|W\|_r = \max_i \frac{(Wr)_i}{r_i} \le \kappa \ell^{\mathsf{T}} W r, \qquad \kappa = \frac{1}{\min_i \ell_i r_i}. \tag{5}$$

Indeed, the scalar on the right before multiplication by κ is a weighted average of these ratios with positive weights $\ell_i r_i$ summing to 1. The induced-norm identity follows from nonnegativity, with equality attained at $x = r$.

For $q \ge 1$ put

$$\varepsilon_q = \sup_{g \ge q} \max_{a,b} \frac{|(B^g - P)_{ab}|}{P_{ab}}. \tag{6}$$

All entries of P are positive. By (2), ε_q is finite, tends to zero, and gives the entrywise bound

$$B^g \le (1 + \varepsilon_q) P \qquad (g \ge q). \tag{7}$$

Lemma 2 (Joining cluster gains)**.** *Let $K^{(1)}, \dots, K^{(s)}$ be finite clusters containing a total of j misses. Then*

$$\prod_{i=1}^{s} \ell^{\mathsf{T}} K^{(i)} r \le S_j.$$

In particular, the product of gains of any collection with at most m total misses is at most M_m.

Proof. Insert h successes between consecutive clusters. For each finite h the result is a j-miss cluster, so its gain is at most S_j. Let $h \to \infty$. Every inserted B^h converges to P, and $PWP = (\ell^{\mathsf{T}} W r)P$ factors the limiting gain. This argument uses a supremum, not attainment of a maximizing gap tuple. □

3.2 A uniform packing bound

Choose an integer $q = q(k)$ with $q \to \infty$ and $q/k \to 0$, for instance $q = \lfloor \sqrt{k} \rfloor$. Take k large enough that $mq + 1 \le k$, and set $L = k - mq > 0$. In a finite prefix of an admissible sequence, join successive misses into the same cluster whenever fewer than q successes lie between them.

Lemma 3 (Window packing)**.** *Each cluster has at most m misses and spans at most mq slots. Assign a cluster to the integer bin $[tL, (t+1)L - 1]$ containing its first miss. All clusters assigned to one bin contain at most m misses in total.*

Proof. If one cluster contained $m + 1$ misses, the span from the first to the last, including both endpoints, would be at most

$$m(q - 1) + (m + 1) = mq + 1 \le k,$$

contradicting the budget. A cluster of at most m misses spans no more than $(m - 1)q + 1 \le mq$ slots.

A cluster starting in bin t ends no later than

$$(t + 1)L - 1 + (mq - 1) = tL + k - 2 < tL + k.$$

Consequently every miss in every cluster assigned to bin t belongs to the same k-window $[tL, tL + k - 1]$. That window has at most m misses. If it extends beyond the chosen prefix, use the original infinite admissible sequence. Truncating the final cluster of a prefix does not increase its span or count. □

Let W be a normalized length-N product and let z be its number of misses. Disjoint full k-blocks and one final fragment give

$$z \le m\lfloor N/k \rfloor + m. \tag{8}$$

The final fragment also has at most m misses because it is contained in a k-window of the infinite extension.

Different clusters are separated by at least q successes. Replace each such success run using (7). Nonnegative matrix multiplication preserves entrywise inequalities. In the scalar $\ell^{\mathsf{T}} W r$, the outer success runs vanish by (2), and the inserted rank-one separators factor the remaining expression into cluster gains. There are at most z separators. By Lemmas 2 and 3, each occupied bin contributes at most M_m. At most $\lceil N/L \rceil$ bins are occupied. Thus, including the zero-miss case,

$$\|W\|_r \le \kappa (1 + \varepsilon_q)^z M_m^{\lceil N/L \rceil}. \tag{9}$$

This is a bound on every admissible word.

Fix k, q first. Taking suprema, Nth roots and the long-product limit superior in (9) yields

$$\log \frac{\rho_{m,k}}{\lambda} \le \frac{\log M_m}{k - mq} + \frac{m}{k} \log(1 + \varepsilon_q). \tag{10}$$

Multiply by k and then let $k \to \infty$. Since $q/k \to 0$ and $\varepsilon_q \to 0$,

$$\limsup_{k \to \infty} k \log \frac{\rho_{m,k}}{\lambda} \le \log M_m. \tag{11}$$

3.3 Matching periodic lower witnesses

Fix any finite cluster K with $j \le m$ misses and length a. For each $k > a$, repeat a length-k word whose normalized period product is $B^{k-a}K$. Every cyclic k-window has exactly j misses. The infinite repetition is admissible, and repeated periods give

$$\frac{\rho_{m,k}}{\lambda} \ge \operatorname{spr}(B^{k-a}K)^{1/k}.$$

As $k \to \infty$, the period product tends to PK. Its nonzero eigenvalue is $\ell^{\mathsf{T}} K r > 0$. Continuity of spectral radius gives

$$\liminf_{k \to \infty} k \log \frac{\rho_{m,k}}{\lambda} \ge \log(\ell^{\mathsf{T}} K r).$$

This holds separately for every fixed finite cluster. Taking the supremum over its gaps, the finite maximum over $j \le m$, and including the all-success word gives the lower bound $\log M_m$. Together with (11), this proves Theorem 1. No exchange of a varying maximizing cluster with the k-limit is used.

4 Finite enclosure of the coefficient

The supremum in (3) can be enclosed by a finite calculation when a Perron tail bound is available.

Proposition 4 (Truncation with limiting gaps)**.** *Suppose $G \geq 1$ and $\varepsilon \geq 0$ satisfy $B^g \leq (1+\varepsilon)P$ entrywise for every $g \geq G$. Let*

$$\mathcal{F}_G = \{I, B, \ldots, B^{G-1}, P\}.$$

Let T_j be the maximum gain obtained by replacing the $j-1$ gap factors in a j-miss cluster by elements of $\mathcal{F}_G$, and put $T = \max(1, T_1, \ldots, T_m)$. Then

$$T \leq M_m \leq (1+\varepsilon)^{m-1} T. \tag{12}$$

Proof. Every occurrence of P is the limit of a success power. Replacing all such occurrences by B^h and taking $h \to \infty$ shows that each finite-search value lies in the closure of genuine cluster values. This proves $T \leq M_m$.

Conversely, in any genuine cluster replace each gap power with exponent at least G by $(1+\varepsilon)P$. There are at most $j-1$ such factors. Positivity gives an upper bound of $(1+\varepsilon)^{j-1}T_j \leq (1+\varepsilon)^{m-1}T$. Take the appropriate suprema and maximum. □

The logarithmic interval has width at most $(m-1)\log(1+\varepsilon)$. Enumeration uses at most $\sum_{j=1}^{m}(G+1)^{j-1}$ labeled cases. Rational data and a rational tail certificate permit exact arithmetic. Obtaining certified Perron data and an all-tail bound for general inputs is a separate task; a floating eigendecomposition or a finite sample of tail powers is not such a certificate. A candidate containing P is generally a limiting witness, not a realizable finite-k product.

5 Gap structure and its limitations

In dimension two, write the second eigenvalue of B as μ, where $|\mu| < 1$. Then $B^g = P + \mu^g(I-P)$, and, with $a = \ell^{\mathsf{T}} E r$ and $b_g = \ell^{\mathsf{T}} E B^g E r$,

$$b_g = a^2 + \beta\mu^g, \qquad \beta = \ell^{\mathsf{T}} E (I-P) E r.$$

The positive and negative cases for μ show that

$$M_2 = \max\{1, a^2, b_0, b_1\}. \tag{13}$$

For $\mu = 0$, the zeroth power is interpreted separately as $B^0 = I$. The finite list in (13) is not dimension independent.

Proposition 5 (Prescribed optimal pair gap)**.** *For every integer $t \geq 1$ there is a strictly positive pair in dimension $n = t+1$ for which b_g is uniquely maximized, over all finite nonnegative gaps, at $g = t$, and its value exceeds the infinite-gap limit.*

Proof. Index coordinates by $0, \ldots, n-1$. Let $R_{i,i+1 \bmod n} = 1$ be the cyclic permutation matrix and let $P = \mathbf{1}\mathbf{1}^{\mathsf{T}}/n$. Set

$$B = \tfrac{1}{2}(R+P), \quad v = \mathbf{1} + ne_0, \quad w = \mathbf{1} + ne_1, \quad E = vw^{\mathsf{T}}/(2n).$$

Take $r = \mathbf{1}$ and $\ell = \mathbf{1}/n$. All entries are positive and $a = 2$. Since $R^g e_0 = e_{-g \bmod n}$ and $B^g = P + 2^{-g}(R^g - P)$, direct multiplication gives

$$b_g = 4 + 2^{-g}\big(n\mathbf{1}_{\{g \equiv t \pmod n\}} - 1\big).$$

Outside the residue class t, this is less than 4. Inside that class it is greater than 4 and strictly decreases with g. The unique maximum is therefore $b_t = 4 + t/2^t > 4$. □

Choosing t outside any proposed fixed finite list defeats that list. This does not exclude dimension-dependent bounds or adaptive truncation, and it is not a statement about exact finite-k maximizing words.

6 A reset application at finite window sizes

The following elementary application of rank-one factorization [1] separates the asymptotic coefficient from finite-window optimality. It is not asserted as an independent novelty claim.

Proposition 6 (Rank-one success mode)**.** *Assume $B = P = r\ell^{\mathsf{T}}$ and set $e_j = \ell^{\mathsf{T}} E^j r$. If $e_1 \le 1$, $e_2 \ge 1$ and $e_3 \le e_2$, then for $k \ge 4$,*

$$\rho_{2,k} = \lambda e_2^{1/k}, \qquad \rho_{3,k} = \lambda e_2^{1/(k-1)}, \qquad M_2 = M_3 = e_2. \tag{14}$$

Proof. Every nonempty success run equals P. Thus $\ell^{\mathsf{T}} W r$ factors into the values e_j of the consecutive miss runs, including boundary runs. Under budget three these runs have length at most three. Delete single misses, and delete one endpoint of each triple. This preserves admissibility and does not decrease the scalar gain, since $e_1 \le 1$ and $e_3 \le e_2$. The remaining runs are pairs. Their starts are at least $k-1$ slots apart, or four misses would fit inside one k-window. At most $N/(k-1)+1$ pairs remain in N slots. Equation (5) proves the upper rate in the second formula. Repeating two misses followed by $k-3$ successes has period $k-1$, is admissible, and has normalized period spectral radius e_2. It attains the matching lower bound.

For budget two, delete single misses. Pair starts must be at least k slots apart: otherwise the last miss of one pair and both misses of the next fit in a k-window. The same bound and a pair repeated every k slots give the first formula. Finally $S_2 = \max(e_2, e_1^2) = e_2$ and $S_3 = \max(e_3, e_1 e_2, e_1^3) \le e_2$, proving the last equality. □

For example, take

$$B = \begin{pmatrix} 99/100 & 1/100 \\ 99/100 & 1/100 \end{pmatrix}, \qquad E = \begin{pmatrix} 1/10 & 1/10 \\ 20 & 1/10 \end{pmatrix}.$$

Here $r = \mathbf{1}$, $\ell^{\mathsf{T}} = (99/100, 1/100)$, and

$$e_1 = 399/1000, \qquad e_2 = 10349/5000, \qquad e_3 = 120797/100000.$$

Set $C = (9/10)B$ and $D = (9/10)E$. Under budget three, uniform exponential stability is equivalent to $\lambda^{k-1} e_2 < 1$. At $k = 7$ and $k = 8$ respectively, the exact comparisons are

$$\frac{5499882909}{5000000000} > 1, \qquad \frac{49498946181}{50000000000} < 1.$$

Thus eight is the first stable window among $k \ge 4$. The maximizing period is $k-1$. When $e_2 > 1$, the two budgets have different finite-k rates despite identical first-order coefficients.

7 Verification and publication status

The accompanying standard-library Python checker uses exact rational arithmetic for prescribed fixtures. It checks 408 admissible-word matrix/packing cases and 7,767 reset run-replacement cases, rejects three deliberately false assertions, and verifies finite instances of the prescribed-gap

identities and coefficient enclosures. These finite checks support the calculations; they do not prove Theorem 1. The all-parameter argument is the classical proof above. The checker is fixture-specific, not a general validator for arbitrary matrix inputs.

The main cluster law is also proved in the accompanying Lean 4.32.2 project. Its audit checks 175 exported declarations at trust level zero, using only the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`. It includes Perron convergence from positivity, the canonical matrix bound, both limiting inequalities, and the norm-rate and gap-supremum identifications. The additional propositions are not all formalized. The manuscript and proof development used AI assistance; kernel checking does not establish publication priority or replace human specialist review. Authorship and affiliations require an explicit decision before circulation. A fresh-context reader review of this manuscript remains pending.

A targeted prior-art audit found no direct match in the inspected theorem statements, but this is not evidence sufficient to establish priority. Important original texts remained unavailable, including Zhao's 2019 work on constrained ergodic optimization for asymptotically additive potentials, Hertneck–Linsenmayer–Allgower's 2021 weakly-hard control paper, and the Gurvits–Samorodnitsky 2005 original attributed in the revision appendix of [1]. No result is attributed directly to an unread text. The external review packet retains these access gaps and the precise comparisons with inspected sources.

References

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